Croup in Children

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Definition

Croup or viral laryngotracheobronchitis (LTB) is one of the frequent causes of stridor. Stridor is a high pitched, harsh sound which occurs during inspiration. Wheeze is a musical sound that occurs during expiration, due to lower airway obstruction.

Epidemiology

Highest incidence among the preschool children (6 month to 3 year of age), occurs during the autumn and winter months. Less than 5% of children with croup, require hospitalization and among those hospitalized only 1–2% require intensive care. Mortality rate in croup is usually <0.5% even for intubated patients.

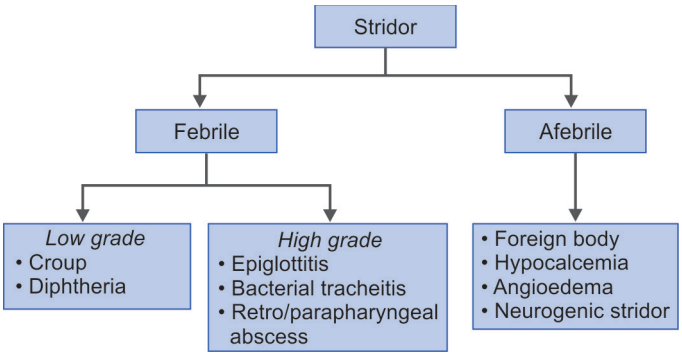
Etiology of Croup

Viral infections: Parainfluenza types 1 and 3 accounts for >70% of viral LTB cases. Other viruses—influenza A, influenza B, adenovirus, respiratory syncytial virus, and metapneumovirus.

Differential Diagnosis

Though croup is the most common cause, there are many other causes of stridor which are given in the **Flowchart 1**.

Flowchart 1: Various causes of stridor and clinical approach.



Diagnosis

Clinical

- ☑ Sudden onset of a distinctive barky cough
- ☑ Usually preceded by upper respiratory infection (URI) symptoms
- ☑ Accompanied by stridor and respiratory distress
- ☑ Hoarse voice

Laboratory

A complete blood count (neutrophilic leukocytosis) and high C-reactive protein (CRP) may help distinguish croup from bacterial etiologies of stridor (e.g., bacterial tracheitis, epiglottitis, peritonsillar abscess, and retropharyngeal abscess), but it is nonspecific.

X-ray Chest

It may show classical steeple sign secondary to glottic and subglottic narrowing **(Fig. 1)**. However, this finding is neither specific nor sensitive for croup. X-ray neck lateral view will be useful in the diagnosis of epiglottitis and retropharyngeal abscess. Identification of organism by doing antigen test or culture can be useful to identify bacterial cause other than croup.

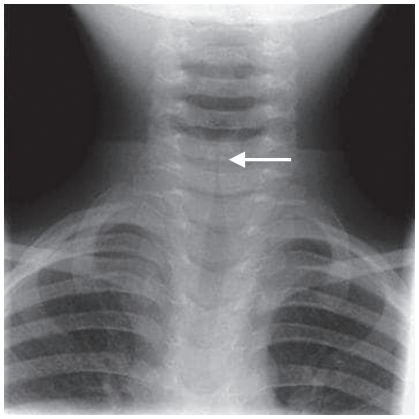


Fig. 1: Church steeple sign in croup (arrow).
Picture Courtesy: Dr Paramarth C

Diagnosis

TABLE 1: Severity assessment.

Signs	*Mild-to-moderate*	*Severe*	*Life-threatening*
Sensorium	Alert (A)	Lethargic arousable (V)	Agitated, pain responsive, or unresponsive (in AVPU scale)
Respiratory distress	No	ICR, SCR, and present	Sternal, SSR, and severe or declining in intensity without improvement in consciousness
Stridor	No stridor at rest	Stridor at rest	Audible stridor becoming quiet without improvement in consciousness and saturation
Heart rate	Normal	Tachycardia	Bradycardia
SpO_2	>95%	>95%	<94%

(AVPU: alert, verbal, pain, unresponsive; ICR: intercostal recession; SCR: subcostal recession; SpO_2: oxygen saturation; SSR: suprasternal recession)

Source: Modified from Bjornson CL, Johnson DW. Croup in children. CMAJ. 2013;185(15):1317-23.

Management

Initial Management

- ☑ Baby should be kept on mother's lap. Separation and crying may worsen stridor.
- ☑ Oxygen should be administered in a nonthreatening manner to maintain oxygen saturation (SpO_2) > 95%.
- ☑ Postpone intravenous access attempt or blood tests, unless it is absolutely needed.
- ☑ Do not insert tongue depressor. If essential, can be done later after stabilization.
- ☑ Do not sedate the child until airway is secured.
- ☑ Never shift the child for X-rays before stabilization.

Specific Management (Flowchart 2)

Mild

- ☑ Oral dexamethasone at 0.6 mg/kg or nebulized budesonide 2 mg.
- ☑ If stable, send home instructing the parents about the natural course and likelihood of recovery in 48–72 hours.
- ☑ Explain the warning signs of worsening (worsening of stridor, poor feeding, or change in level of consciousness) and instruct to come to emergency department (ED), if worsening happens.
- ☑ There is no role for antibiotics or beta-agonist nebulization in viral croup.

Moderate-to-severe

- ☑ It is preferable to hospitalize the child.
- ☑ *Adrenaline nebulization:*
  - Undiluted 1:1,000 adrenaline 0.5 mL/kg is mixed with normal saline (NS) to a maximum dose of 5 mL. For example, in a child with body weight 8 kg, adrenaline 4 mL and 1 mL of NS is used. In a child with 10 kg and beyond undiluted adrenaline can be used without NS.

Management

- Second dose can be repeated after 2 hours, if needed.
- *Caution*: Here epinephrine is used as nebulization and not as parenteral route.

☑ *Steroids:*

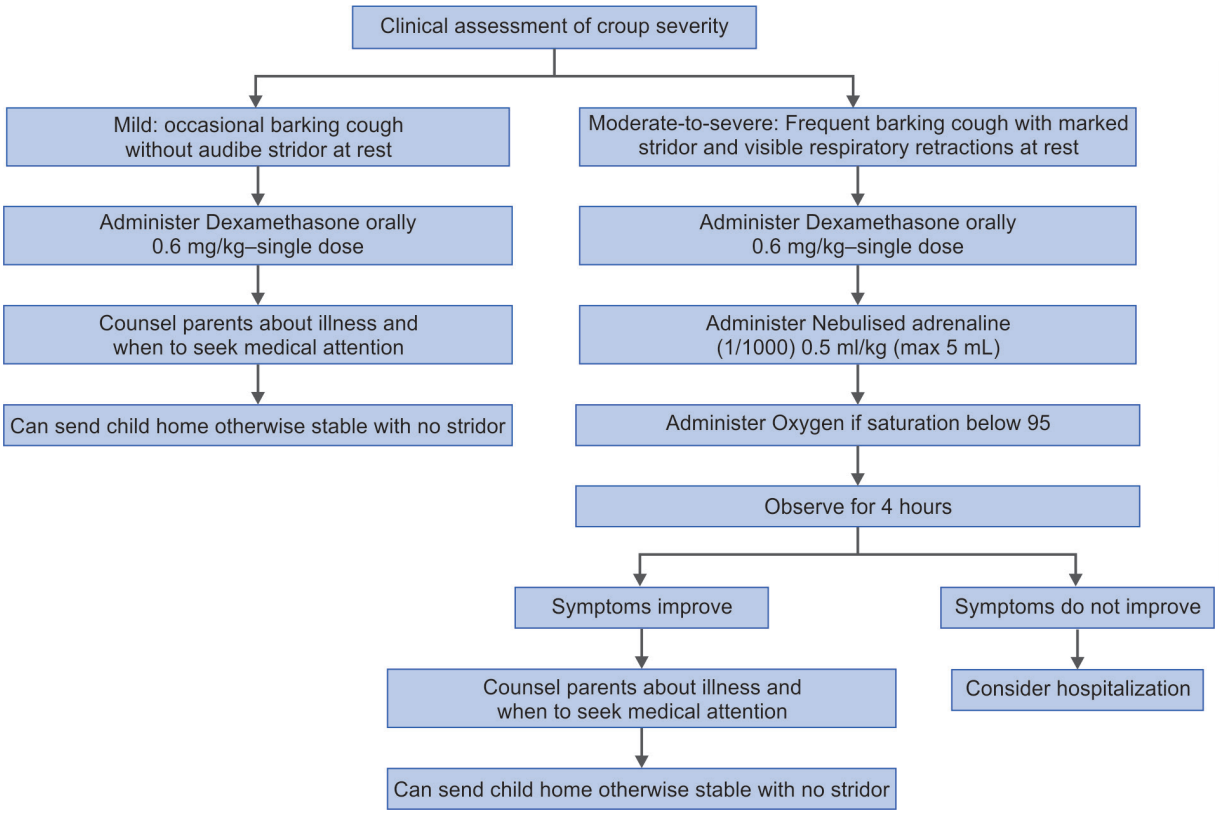
- It is indicated even in children who recovers after adrenaline nebulization as the effect of adrenaline will wane after 2 hours.
- Dexamethasone—0.6 mg/kg (maximum—8 mg) oral or IV or IM
- Nebulized budesonide 2 mg (dose same at all ages)
- Majority may require single dose of steroids, but in severe cases frequent doses may be needed for 48 hours.

Indication for Immediate Referral to Hospital

☑ If the child is pain responsive or unresponsive, having reduced respiratory effort, and saturation <94%.

☑ Accompany the child, simultaneously supporting with bag valve ventilation with oxygen.

Flowchart 2: Algorithm for management of croup.



Source: Zoorob R, Sidani M, Murray J. Croup: an overview. Am Fam Physician. 2011;83(9):1071.

Further Reading

- ☑ Bjornson CL, Johnson DW. Croup in children. CMAJ. 2013;185(15):1317-23.
- ☑ Rodrigues K, Roosevelt G. Acute inflammatory upper airway obstruction (croup, epiglottitis, laryngitis, and bacterial tracheitis). In: Kligeman R, St Geme III J (Eds). Nelson Textbook of Pediatrics, 21st edition. Philadelphia: Elsevier; 2020. pp. 2202-5.
- ☑ Wilmott RW, Deterding RR, Li A, Ratjen F, Sly P, Zar H, et al. Kendig's Disorders of the Respiratory Tract in Children, 9th edition. Amsterdam, Netherlands: Elsevier; 2019. pp. 406-12.
- ☑ Zoorob R, Sidani M, Murray J. Croup: an overview. Am Fam Physician. 2011;83(9):1071.

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Community-Acquired Pneumonia

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